ERPsimLab
HEC MONTRÉAL

ERPsim for SAP HANA

ERPsim Sustainability Carbon Footprint Reporting

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Corporate Carbon Footprint and GHG Protocol

Tracking and reporting carbon emissions is the foundation of any sound sustainable strategy. An efficient reporting of a company's greenhouse gas emissions is the cornerstone of understanding the impact of the company's activities on their environment and in identifying opportunities to reduce their footprint.

The Greenhouse Gas Protocol, also called GHG Protocol, is a joint initiative of World Resources Institute (WRI) and World Business Council for Sustainable Development (WBCSD) to establish "comprehensive global standardized frameworks to measure and manage greenhouse gas (GHG) emissions from private and public sector operations, value chains and mitigation actions."[1]

GHG Protocol provides multiple standards for business and governments to report carbon emissions. The following documentation is based on the Corporate Accounting and Reporting Standard created by GHG Protocol for companies reporting corporate-level emissions.

Carbon Footprint Reporting with ERPsim

In the ERPsim Sustainability scenario of the Logistics and Manufacturing games, most companies' activities will generate carbon emissions in kilograms of carbon dioxide equivalent (CO_2e). As companies will be charged a carbon tax for their carbon emissions, each kilogram of CO_2e generated will have a negative impact on the companies' profit. Furthermore, customers are more interested in purchasing products from a sustainable company, which translate in a higher perceived utility for products sold by companies with a low carbon footprint. This will not translate into more units sold but will allow sustainable companies to increase their margins accordingly.

Using the OData view *Carbon_Emissions*, students can analyze their CO_2e emissions data to identify their different sources of emissions and adjust their strategies to minimize their carbon footprint while still trying to maximize their profits.

Following the Greenhouse Gas Protocol Corporate Standard guidelines, all the simulation companies' carbon emissions are categorized in 3 distinct groups: Scope 1, Scope 2, and Scope 3.

[1] Greenhouse Gas Protocol. (n.d.). Retrieved 20 January 2023 from https://ghgprotocol.org/about-us

Scope 1 – Direct Emissions

Scope 1 contains all direct CO2e emissions from a company's owned or controlled sources. This includes, but is not limited to, emissions from the production or transformation of products, transportation from controlled freight fleet, and fugitive emissions.

In the ERPsim Logistics Sustainability scenario, four (4) carbon-generating activities are grouped in Scope 1. In Manufacturing Sustainability, Scope 1 is composed of five (5) carbon-generating activities:

1. **Goods Movement Between Supplier and Main Warehouse**

   As the reporting company is collecting purchase orders from the simulation suppliers' warehouses using their own freight fleet, carbon emissions from these goods receipts are under the control of the reporting company and are therefore reported as Scope 1.

   Students' supplier choice and procurement frequencies will have a direct impact on this source of carbon emissions.

2. **Goods Movement Between Main Warehouses and Customers**

   All sale orders that are being delivered from your main warehouse to your customers will be generating CO2e emissions. As these emissions are generated by the trucks owned by the reporting company and used to deliver the products to the customers' homes, they are considered as direct emissions from owned combustion sources and are accordingly grouped in Scope 1.

   Opting out of selling from the main warehouse and/or using regional warehouses is the main decision students could take to increase or reduce this source of carbon emissions. Furthermore, the students' pricing strategies can also have a significant impact on these emissions as a lower quantity of sales orders will translate in fewer deliveries, and thus less carbon emissions.

3. **Goods Movement Between Main Warehouse and Regional Warehouses**

   All internal goods transfers between a company's main warehouse and their regional warehouses are done using the company's own freight fleet and will consequently generate CO2e emissions reported in Scope 1.

   Students must optimize their internal transfers strategies to minimize these carbon emissions. Moreover, opting in selling from the main warehouse can also reduce the number of internal transfers, and therefore this source of emissions.

4. Overstocking

When a company is in an overstocking situation, they are effectively renting refrigerated containers (reefers) from a specialized partner to temporarily store the exceeding stocks in a suitable environment. As the reefers are under the reporting company's control while they are in use, CO2e emissions from the reefers' cooling systems will be reported in Scope 1.

Students are not able to reduce the carbon emissions from rented reefers and must therefore minimize the time spent being overstocked to reduce this source of emissions.

5. Production and Setup time (Manufacturing only)

In Manufacturing Sustainability, all production activities, including the setup time between production orders, generate carbon emissions from the machinery usage and operation. As they are fully under the control of the reporting company, these emissions are reported in Scope 1.

Students must optimize their production orders schedule to minimize the time spent in setup time. Moreover, investing in sustainable manufacturing will have a significant impact on the carbon emissions generated by production activities.

Scope 2 – Energy Indirect Emissions

Scope 2 is used to group all indirect emissions from purchased energy (e.g., electricity, heat) consumed by the reporting company but generated and provided by a utility company or any uncontrolled sources.

Although this category is quite restrictive, it usually accounts for one of the biggest sources of CO2e emissions in most real-life companies.

In the ERPsim Logistics Sustainability and Manufacturing Sustainability scenarios, only one (1) carbon-generating activity is included in Scope 2:

1. Overhead – Purchased Energy

Part of the reporting company's overhead CO2e emissions comes from the energy purchased from the utility company to provide electricity to the company's office and multiple warehouses. Following the GHG Protocol guidelines, all overhead emissions from purchased energy are reported in Scope 2.

While Scope 2 is usually one of the biggest sources of CO2e emissions in real-life businesses, it accounts only for a small portion of carbon emissions in the simulation companies in ERPsim, as students do not currently have any control on the overhead emissions of their companies. Therefore, we opted to reduce the significance of Scope 2 reporting in the simulation in favor of increasing the weight of Scope 1 and Scope 3, as students have more control over these groups of emissions.

Scope 3 – Other Indirect Emissions

Scope 3, which is considered optional to report, groups all other indirect upstream and downstream emissions having an impact on your value chain. These emissions are generally owned and reported as Scope 1 by another party, but it is still highly valuable for a company to track and report their Scope 3 emissions as it could give valuable insight on your partners' carbon footprints or your products' usage and end-of-life carbon emissions.

While a reporting company has less control over their Scope 3 emissions than the other groups, they can still use the data to revaluate their value chain and identify cost saving or emission-reducing opportunities.

In the ERPsim Logistics Sustainability scenario, three (3) carbon-generating activity is included in Scope 3. In Manufacturing Sustainability, Scope 3 is composed of four (4) carbon-generating activities:

1. **Products Purchased from Suppliers**

   Every unit of any product purchased from a supplier will come with a carbon footprint from the suppliers' operations. As these carbon emissions come from the suppliers' direct operations, they are not owned or controlled by the reporting company and are therefore grouped in Scope 3.

   Students' control over this type of emission is limited to the supplier choice and quantity ordered during the planning and procurement processes.

2. **Goods Movement Between Regional Warehouses and Customers**

   When selling from regional warehouses, customers are effectively driving and picking up products from their closest regional warehouse. As the transportation carbon emissions are generated by the customers' own car, thus not owned, or controlled by the reporting company, they are reported in Scope 3.

   Opting in selling from the main warehouse and students' pricing strategy will have an impact on these emissions, as selling exclusively from the main warehouse and a lower quantity of sales orders will translate in less picking from regional warehouses, and therefore less carbon emissions.

3. **Overhead – Others**

   The remaining overhead carbon emissions from the reporting company are grouped together in Scope 3 and represent employees commutes and waste disposal.

   Students do not currently have any control on the overhead emissions of their companies and every company has the same overhead carbon emissions for the whole duration of the simulation.

4. Investments

In Manufacturing Sustainability, reporting companies can invest in different regular or sustainable investment options, which will generate carbon emissions. The emissions from these investments come from the upstream carbon emissions of energy-efficient machinery and trucks purchased by the reporting company, as well as carbon emissions from consultants' visits and commuting.

Students must evaluate the trade-off between the carbon emissions and the benefits of the investments and decide which investment(s) better suit their strategy.

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